

Rotation Matrix in 3D — from 2D, via the Z-axis first

1 Warm-up: 2D rotation *is* rotation about the z -axis

In 2D we derived the rotation matrix (counterclockwise by θ):

$$R_{2D}(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Now observe: a rotation *in the $\mathbb{R}^2$ plane* is exactly a rotation in $\mathbb{R}^3$ *about the z -axis*, where every point keeps its z -coordinate fixed. The axis of the 2D rotation is therefore the z -axis. This gives us the 2D matrix *embedded* into 3×3 : keep the 2×2 block, and append the z -axis entries so that z is untouched:

$$R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Multiplying a point (x, y, z) :

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \\ z \end{pmatrix}.$$

So the x, y part spins by the 2D matrix, and $z' = z$: the **z -component is untouched**. This is the whole content of “2D rotation = rotation about z .”

Note: the z -axis is not “imaginary” — it is the perfectly real third direction that is simply *perpendicular to the page* in a 2D drawing. It points “out of the paper.”

2 The same recipe for all three axes

By the identical reasoning, rotation about y (angle ϕ) and about x (angle ψ) are got by keeping that axis’s row and column as the axis itself and putting the 2D block in the other two coordinates:

$$R_y(\phi) = \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix}, \quad R_x(\psi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & -\sin \psi \\ 0 & \sin \psi & \cos \psi \end{pmatrix}.$$

In each matrix the **pivot axis’s row and column** read like the axis vector itself (1 on the diagonal, 0 elsewhere), and the remaining 2×2 block is exactly the 2D rotation within the perpendicular plane. (The sign placement in R_y differs from R_z because the x - z plane is read with y pointing the other way; both blocks are ordinary 2D rotations in their planes.)

3 What are roll, pitch and yaw? (the human frame)

In the usual “body” frame for a human/aircraft: x = forward (nose), y = out the right, z = down.

- **Yaw** — rotation about the z -axis (vertical). Turning the heading/head left–right. Matrix $R_z(\theta)$, in the xy -plane.

- **Pitch** — rotation about the y -axis (lateral). Nose up/down (climb/dive). Matrix $R_y(\phi)$, in the zx -plane.
- **Roll** — rotation about the x -axis (forward). Bank/tilt left (or right). Matrix $R_x(\psi)$, in the yz -plane.

The ZYX intrinsic order = yaw first, then pitch, then roll: point the nose, raise it, bank it.

4 The easy way to derive R_x and R_y : re-run 2D in each other plane

You do *not* re-derive anything new — you reuse the 2D result, just with different letters playing “ x and y .” The recipe:

Pick the two axes of the plane, apply the 2D formula with them in their natural order, then rewrite the three equations in (x, y, z) order. The pivot axis gets a $(0 \dots 1 \dots 0)$ row/column; everything else is the 2D block, laid in by coordinate.

R_x = **rotate the yz -plane** (pivot x , untouched). The plane’s two axes are y and z , and $\{y, z\}$ already reads in coordinate order, so the 2D formula applies directly:

$$y' = y \cos \psi - z \sin \psi, \quad z' = y \sin \psi + z \cos \psi, \quad x' = x,$$

$$R_x(\psi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & -\sin \psi \\ 0 & \sin \psi & \cos \psi \end{pmatrix}.$$

The 2D block is literally the 2D matrix we derived, at the yz positions; the x row/column is the axis itself.

R_y = **rotate the zx -plane** (pivot y , untouched). Here the plane’s natural order is $\{z, x\}$ (so “ z ” plays the role of the first coordinate and “ x ” the second). Apply the 2D formula to z and x :

$$z' = z \cos \phi - x \sin \phi, \quad x' = z \sin \phi + x \cos \phi, \quad y' = y.$$

Now rewrite in (x, y, z) output order (this reordering is what flips the signs):

$$x' = x \cos \phi + z \sin \phi, \quad y' = y, \quad z' = -x \sin \phi + z \cos \phi,$$

$$R_y(\phi) = \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix}.$$

Why it is not *just* “mixing up the orders.” Two things happen, and you need both: (1) you paste the 2D block into the right positions and put (1) on the pivot diagonal (*this is the “mix-up-orders” part*), and (2) you must re-read the plane’s coordinates in (x, y, z) order. In the yz -plane, $\{y, z\}$ already reads in order, so the signs come out exactly like 2D. In the zx -plane the natural order is $\{z, x\}$ but your vector is written (x, y, z) ; that reordering flips the sign into the corners. So the sign flip in R_y is *not arbitrary* — it is the bookkeeping of expressing “ z first, x second” in (x, y, z) coordinates.

5 Order matters: the ZYX (intrinsic) Euler convention

A *general* 3D orientation is a composition of these single-axis rotations, and **order matters**: rotating about z then about y is not the same as the reverse. The standard aerospace / robotics convention is **intrinsic Z–Y–X** (body-axis “yaw–pitch–roll”):

1. **Yaw**: rotate about the body’s z -axis by θ .
2. **Pitch**: rotate about the body’s *new* y -axis by ϕ (the y -axis has *moved* after step 1).

3. **Roll:** rotate about the body's *new* x -axis by ψ (moved again after step 2).

Each subsequent rotation happens about the axis that has just been rotated — this is the “body frame” behavior shown in the video. The total rotation is the right-to-left product

$$R = R_x(\psi) R_y(\phi) R_z(\theta)$$

which is applied to a vector as $v' = Rv$; the rightmost matrix R_z acts first, then R_y on its result, then R_x , matching the intrinsic order.

Fixed vs rotating axes (the subtlety). If instead one rotates about the *fixed* world axes, the composite matrix comes out *reversed*:

$$\text{intrinsic } z, y, x \quad \Longleftrightarrow \quad \text{extrinsic } x, y, z.$$

That is, rotating about the moving y -then- x equals rotating about the fixed x -then- y in reverse order. The two readings describe the same final geometry; only the order of ψ, ϕ, θ in the product interpretation changes.

6 Summary

- 3D rotation about any single axis = the 2D rotation in the perpendicular plane, with that axis left untouched.
- R_z is the very 2D matrix we derived, plus the 0,0,1 row/column.
- A general 3D orientation is an *ordered* product of single-axis ones; the ZYX intrinsic convention applies z , then y , then x , about the already-rotated axes.